SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
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॥वसुधैव कुटुम्बकम्॥

CSE - Academic Year 2026-27
Java Programming Lab | Batch 2025-29

ASSIGNMENT : 21

Name of Student	Anushree Shinde
PRN No.	25070122041
Batch	CSE A - 2
Class	Java Programming Lab
Academic Year & Semester	2025-29 : Semester 3
Date of Submission	

TITLE : Demonstration and exploration of JDBC classes and interfaces (DriverManager, Connection, Statement).

Learning Objectives :

To understand and use the core JDBC classes for database connectivity.

Theory :

The **JDBC API** provides classes and interfaces that allow Java applications to connect and communicate with databases. **DriverManager** is a JDBC class used to establish a connection with a database using the database URL, username, and password. The **Connection** interface represents the connection between the Java application and the database and provides methods to create statements. The **Statement** interface is used to execute SQL statements and retrieve or modify database records. These components work together to establish communication between a Java application and a database.

Exercise :

1. Create a program to connect Java with database and display database connection status.

```
import java.sql.*;

public class java21a {
    public static void main(String[] args) {
        try {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/anushree",
                "root",
                "sit@123"
            );

            System.out.println("Database connected successfully!");
            System.out.println("Connection is active.");

            con.close();

        } catch (Exception e) {
            System.out.println("Database connection failed!");
            System.out.println(e);
        }
    }
}
```

OUTPUT :

```
Database connected successfully!
Connection is active.
```

With Error :

```
Database connection failed!
java.sql.SQLSyntaxErrorException: Unknown database 'anuhree'
```

2. Create a Java program to connect with a database and display the **connection status**. After successful connection, show a message indicating that the **Student database is connected successfully**.

```
import java.sql.*;

public class java21b {
    public static void main(String[] args) {
        try {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/anushree",
                "root",
                "sit@123"
            );

            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("select database()");
            if (rs.next()) {
                System.out.println("Student database is connected
successfully.");
                System.out.println("Connected Database: " +
rs.getString(1));
            }
            con.close();

        } catch (Exception e) {
            System.out.println("Database connection failed!");
            System.out.println(e);
        }
    }
}
```

OUTPUT :

```
Student database is connected successfully.
Connected Database: anushree
```

With Error :

```
Database connection failed!
java.sql.SQLSyntaxErrorException: Unknown database 'anshree'
```

Outcome :

- Understood the basic JDBC components used for database connectivity.
- Learned the use of **DriverManager** to establish a database connection.
- Understood the role of the **Connection** interface in JDBC.
- Learned the purpose of the **Statement** interface for executing SQL statements.
- Successfully established a connection between a Java application and a MySQL database.

Conclusion :

This assignment demonstrated the basic JDBC components **DriverManager, Connection, and Statement** used for database connectivity. A Java application was successfully connected to the MySQL database and the connection status was displayed. The assignment helped in understanding the basic process of establishing communication between a Java application and a database.